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Learning bilingual word embedding for automatic text summarization in low resource language

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ABSTRACT

Studies in low-resource languages have become more challenging with the increasing volume of texts in today's digital era. Also, the lack of labeled data and text processing libraries in a language further widens the research gap between high and low-resource languages, such as English and Indonesian. This has led to the use of a transfer learning approach, which applies pre-trained models to solve similar problems, even in different languages by using bilingual or cross-lingual word embedding. Therefore, this study aims to investigate two bilingual word embedding methods, namely VecMap and BiVec, for Indonesian – English language and evaluates them for bilingual lexicon induction and text summarization tasks. The generated bilingual embedding was compared with MUSE (Multilingual Unsupervised and Supervised Embeddings) as the existing multilingual word created with the generative adversarial network method. Furthermore, the VecMap was improved by creating shared vocabulary spaces and mapping the unshared ones between languages. The result showed the embedding produced by the joint methods of BiVec, performed better in intrinsic evaluation, especially with CSLS (Cross-Domain Similarity Local Scaling) retrieval. Meanwhile, the improved VecMap outperformed the regular type by 16.6% without surpassing the BiVec evaluation score. These methods enabled model transfer between languages when applied to cross-lingual-based text summarization. Moreover, the ROUGE score outperformed classical text summarization by adding only 10% of the training dataset of the target language.

1. Introduction

The growth of the internet has resulted in a data explosion, which continues to generate new tasks and domains. Moreover, new annotations are certainly time-consuming and costly. This problem has enlarged the research gap between low and high-resource languages. The recent advances in NLP study have led to the use of a transfer learning approach for these new tasks and domains by leveraging pre-trained NLP models.

Transfer learning consists of an inductive and transductive setting. Inductive transfer learning is when the source task is different from the target. This requires labeled data in the target task, which is difficult to obtain in low-resource languages. Furthermore, its size and quality tend to influence the results obtained. On the other hand, transductive has the same source and target tasks, but both have different feature spaces, such as domains or languages. Cross-lingual transfer learning (CLTL), as a transductive method, is the most feasible way to address issues in low-resource languages since it only requires labeled data in the source task. Several available resources in the source language are used to solve similar tasks in that of the target, either with zero-shot learning or fine-tuned with only a few datasets. This approach is feasible because several languages share similar lexical, syntactic, and semantic structures.

CLTL requires cross-lingual word embedding (CLWE) to enable transfer learning to capture shared representation in both languages (Ruder et al., 2019). Several CLWE models in natural language processing have been developed, particularly for English, but Indonesian did not get that much interest. The only static CLWE currently available for Indonesian is MUSE (Conneau

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et al., 2018) which was developed on an adversarial basis. Therefore, this study aims to investigate other approaches and make an improvement to obtain a more robust CLWE. Since it only creates a shared space for two languages, namely English and Indonesian, we henceforth use the term bilingual word embedding (BWE).

Currently, there are two main approaches for generating cross-lingual/bilingual word embedding: the mapping and the joint methods (Ormazabal et al., 2019). The mapping method requires the monolingual corpus to train each language's embeddings individually before mapping both into a shared vector space using a linear transformation. Meanwhile, the joint method builds word representations from multiple languages in a parallel corpus simultaneously. VecMap (Artetxe et al., 2018) and MUSE are two well-known techniques that use the mapping method with orthogonal transformations. According to (Doval et al., 2020), VecMap is more robust than MUSE, which does not perform well with small dictionaries and has some stability issues. However, a fully unsupervised VecMap was built on the assumption that monolingual embedding spaces are approximately isomorphic and this assumption is too restrictive, as it is not always satisfied (Søgaard et al., 2018). Therefore, this study will enhance the unsupervised alignment method by utilizing the shared space concept to produce a better BWE.

Several studies have adopted cross-lingual transfer learning in many NLP tasks, such as machine translation, sentiment analysis, and dependency parsing, named entity recognition (Pikuliak et al., 2021). However, few implementations of text summarization tasks have not yielded strong performance (Peng et al., 2021; Žagar and Robnik-Šikonja, 2020). This challenge encourages the study of cross-lingual transfer learning (CLTL) for the task of text summarization, including the use of BWE in text summarization architecture.

The BWE learning model for English-Indonesian was initially investigated by comparing the mapping and joint methods under similar conditions. VecMap (Artetxe et al., 2018), as a mapping method, and BiVec (Luong et al., 2015), as a joint method, were selected because they are both popular in generating BWE in other language pairs (Marie and Fujita, 2019; Ormazabal et al., 2019; Sabet et al., 2019). They also generate separately aligned word embeddings for each language, making them easier to implement in downstream tasks. The alignment in VecMap was enhanced to have a better or nearly equal performance as BiVec.

The bilingual vector representation, resulting from VecMap and Bivec, tends to be evaluated both intrinsically and extrinsically. Bilingual lexicon induction is used for intrinsic evaluation, while extrinsic assessment is carried out by applying a cross-lingual approach to text summarization perceived as an NLP downstream task. Training of text summarization in English was performed, and the model was further used to summarize the Indonesian document. The study on the use of pre-trained model has become more popular. However, to the best of the author's knowledge, none has been carried out on cross-lingual transfer learning in the Indonesian text summarization task. The present study folds in three contributions: (1) the investigation of VecMap and BiVec as feasible learning methods of the bilingual word embedding for English – Indonesian (2) the improvement of VecMap by applying unsupervised joint training and aligning only unshared vocabulary (3) the application of bilingual word embedding in a cross-lingual approach for text summarization task.

The first section introduces the study, while the second section analyzes preliminary related works, explains and compares the bilingual word embedding and summarization methods for Indonesian text. The section III methodology is followed by the section IV experiment and evaluation. It is completed with Section V, which contains the conclusion.

2. Related works

Bilingual Word Embedding (BWE), defined as mapping word representations from one language to another, is the key to enabling cross-lingual transfer learning. This section analyzes some of its learning approaches and summarization tasks in Indonesian. In addition, the summarization architecture used to carry out cross-lingual transfer between English and Indonesian is briefly explained in this study.

2.1. Cross-lingual word embedding

Typically, a dictionary is needed for language translation. However, the linear mapping of continuous word representations in both source and target languages is currently being utilized. Two methods are commonly employed in BWE development: mapping method and joint method (Ormazabal et al., 2019). The mapping method employed a monolingual corpus to train word embeddings in each language. It further maps two embeddings into a vector space using a linear transformation. Meanwhile, the joint method simultaneously extracts word representation from several languages on parallel or comparable corpora.

The mapping method was carried out by (Mikolov et al., 2013), who identified a linear function from one embedding to another by minimizing the distance between the mapping and the actual vectors. However, this approach requires a seed of bilingual lexicon, spanning at most several thousand-word pairs, making it less efficient in languages with limited linguistic resources. Recent studies have used topological similarities between monolingual vector spaces to develop fully unsupervised methods for identifying linear transformations that do not require bilingual supervision, such as adversarial approach, self-learning approach, etc.

The adversarial approach uses Generative Adversarial Network (GAN) architecture (Goodfellow et al., 2014), composed of a generator and a discriminator. The generator creates the data as well as possible to fool the discriminator. Therefore, linear mapping tends to take the generator's weight as its output. This method was first proposed by (Zhang et al., 2017), and several improvements were later made by (Conneau et al., 2018). This includes the refinement process which keeps the most frequent word as an anchor for linear transformations and measuring word similarity with cross-domain similarity local scaling (CSLS). This GAN-based alignment method results in multilingual word embedding for several languages aligned in a single vector space as well as a bilingual dictionary evaluation dataset, known as MUSE.

The self-learning approach proposed by (Artetxe et al., 2018) has a different objective from the previous approaches. It aims to find the shared space of the two languages rather than discovering the linear function. The two embeddings are subsequently mapped in the shared space by their respective functions. This method is more realistic and generalizable, even on distant language pairs. However, it requires unsupervised initialization, which remains a challenge to date. This led to the investigation of a self-learning approach known as VecMap, which does not require extensive computational resources and is feasibly implemented in resource-limited languages.

VecMap consists of four sequential steps: embedding normalization, unsupervised initialization, iterative self-learning, and symmetric reweighting. The first starts with normalization based on length, followed by the mean centers of each dimension, and length-based normalization again. This step aims to ensure that the dot product of any two embeddings is equivalent to their cosine similarity. Afterward, the unsupervised initialization of the dictionary is carried out based on the strong assumption that two equivalent words in different languages should have a similar distribution of similarity. This second step is the biggest challenge since building an initial solution without any parallel data or a seed dictionary is hard to be the nearest possible to the optimal solution. Random initialization would lead the algorithm to bad performances because it tends to get stuck in poor local optima. Therefore, (Artetxe et al., 2018) proposed some key improvements to make it more robust and perform better mappings. This includes a stochastic dictionary induction (randomly keeping some values in the dictionary), frequency-based vocabulary cutoff (keeping only the k most frequent words in each language), CSLS retrieval (penalize the similarity score of hubs), as well as bidirectional dictionary induction (the concatenation dictionary in both directions, from source to target languages and vice versa). After processing these improvements, self-learning was iteratively run until it converged. First, it maximizes the similarities of the two mappings with the dictionary, which was further updated. In the last step, reweighting tends to change the values to improve the dictionary quality slightly.

On the contrary, the joint method functions by aligning the corpus, either at the word (Gouws and Søgaard, 2015; Luong et al., 2015), sentence (Gouws et al., 2015; Sabet et al., 2019), or document levels (Vuli and Moens, 2016). A recent strategy was embedded at the sub-word level to extend the vocabulary in the shared space across languages. (Lample and Conneau, 2019) designed the Cross-lingual Language Model (XLM) as a multilingual extension of BERT (Devlin et al., 2019). It generated sub-word embedding using the Byte Pair Encoding (BPE) approach, as well as dual-language training mechanisms to explore word relationships in different languages. Although this approach is used to obtain significant results when evaluated with a zero-shot sentence entailment task, the training process requires large computational resources.

As a result, it was investigated by comparing the BiVec (a.k.a. BiSkip), and VecMap approaches proposed by Luong et al. (2015) and (Artetxe et al., 2018), respectively. Furthermore, several studies have used these two approaches as a baseline to create cross-lingual word embedding (Marie and Fujita, 2019; Ormazabal et al., 2019; Sabet et al., 2019). This motivated the present study to compare their experimental findings in language pairs other than Indonesian-English.

BiVec is a simple extension of the SkipGram model used to resolve a multilingual problem. However, instead of predicting the surrounding words in the source language, it is also able to predict parallel words in the target language. Moreover, since BiVec training is directly parallel to the corpus, no mapping function is required. Instead, it exploits monolingual and bilingual contexts within a shared space. The essence is to optimize monolingual and cross-lingual loss, as shown in the following equation:

$$\alpha(Language_1 + Language_2) + \beta Bilingual \tag{1}$$

The monolingual model captures the structure of each language. Meanwhile, the bilingual component binds the two monolingual spaces together. It involves using the parameters α and β to balance the influence of monolingual components on bilinguals.

2.2. Text summarization methods

Even though Indonesian is spoken by approximately 200 million people and is the sixth most language used on the web[1], its research progress is still under-represented in NLP study due to lack of dataset and linguistic resources. Several studies use personal datasets, making it difficult to compare their results to others. Some have attempted to create a public summarization corpus for the Indonesian language since 2016. For example, (Koto, 2016) developed a corpus from WhatsApp conversations, (Kurniawan and Louvan, 2018) and (Koto et al., 2020) created datasets from online news. Even though the availability of this large dataset allows the use of deep neural networks in Indonesian text summarization tasks, only a few studies have been conducted to that effect. Before 2019, most methods used in summarizing Indonesian texts were unsupervised and non-neural networks. Therefore, some studies focus on optimizing the features (Budhi et al., 2007; Gunawan et al., 2017; Herdiyeni et al., 2012; Maryana et al., 2018; Sabuna and Setyohadi, 2017; Silvia et al., 2014; Tardan et al., 2013). Summarization based on the deep neural network has adopted the benchmark dataset (Cai et al., 2019; Chandraseta and Khodra, 2019; Halim et al., 2020; Wijayanti et al., 2021).

This study uses NeuralSum (Cheng and Lapata, 2016) to carry out the cross-lingual approach for extractive text summarization, and its architecture is shown in Fig. 1. This method is developed using the encoder-decoder architecture, with CNN and LSTM as encoders and document readers. CNN generates sentence embedding and subsequently forwards it to LSTM to produce a document representation capable of capturing local as well as global information. Meanwhile, the decoder uses another LSTM to label the sentence sequentially, considering its relevance and redundancy with the others. This process involves the use of both document representation and labeled sentences in the previous position.

Given a document sentence s consisting of a sequence of n words $(w_1, \cdots, w_n)$ represented by $W \in \mathbb{R}^{n \times d}$ with d as embedding dimension. While CNN extractor has multiple kernels $K \in \mathbb{R}^{c \times d}$ as feature maps with different window size c, the convolution applied as follows (Cheng and Lapata, 2016):

$$\mathrm{f} = \tanh(W \otimes K + b) \tag{2}$$

where b is bias. Furthermore, max pooling is performed to obtain a single feature of sentence in each kernel width. Since this process produces multiple sentence vectors, the final sentence representation is obtained by summing these sentence vectors. Once sentence representations are obtained by using the CNN extractor, they are fed into LSTM encoder as document reader.

Meanwhile, a document D consisting of m sentence vectors generated from CNN $(s_1, \cdots, s_m)$, the hidden state at time step t, denoted by h_t, is updated as (Cheng and Lapata, 2016):

$$\begin{bmatrix} i_t \\ f_t \\ o_t \\ \hat{c}_t \end{bmatrix} = \begin{bmatrix} \sigma \\ \sigma \\ \sigma \\ tanh \end{bmatrix} W \cdot \begin{bmatrix} h_{t-1} \\ s_t \end{bmatrix} \tag{3}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \hat{c}_t \tag{4}$$

$$h_t = o_t \odot \tanh(c_t) \tag{5}$$

where W is learnable matrix weights. The LSTM's hidden states $(h_1, \cdots, h_m)$ are subsequently used as the final sentence representations.

Furthermore, sentence extraction is conducted on the decoder using another LSTM. It utilizes the attention mechanism, which is applied to extract the most important sentence after reading them. Given sentence vector $(s_1, \cdots, s_m)$ and encoder's hidden states from LSTM $(h_1, \cdots, h_m)$, while decoder's hidden states $\left(\bar{h}_1, \cdots, \bar{h}_m\right)$ at time step t is calculated by the equation (Cheng and Lapata, 2016):

$$\bar{h}_t = LSTM\left(p_{t-1}s_{t-1}, \bar{h}_{t-1}\right) \tag{6}$$

where p_{t-1} is the decoder probability to extract the previous sentence, which is regarded as an important sentence. The p_{t-1} label is initially set to true, and its value gradually shifts to the predicted label during training.

The binary decision is modelled by the sigmoid layer as follows (Cheng and Lapata, 2016):

$$p(y(t) = 1|D) = \sigma\left(MLP\left(\bar{h}_t : h_t\right)\right) \tag{7}$$

[1] https://www.internetworldstats.com/stats7.htm.

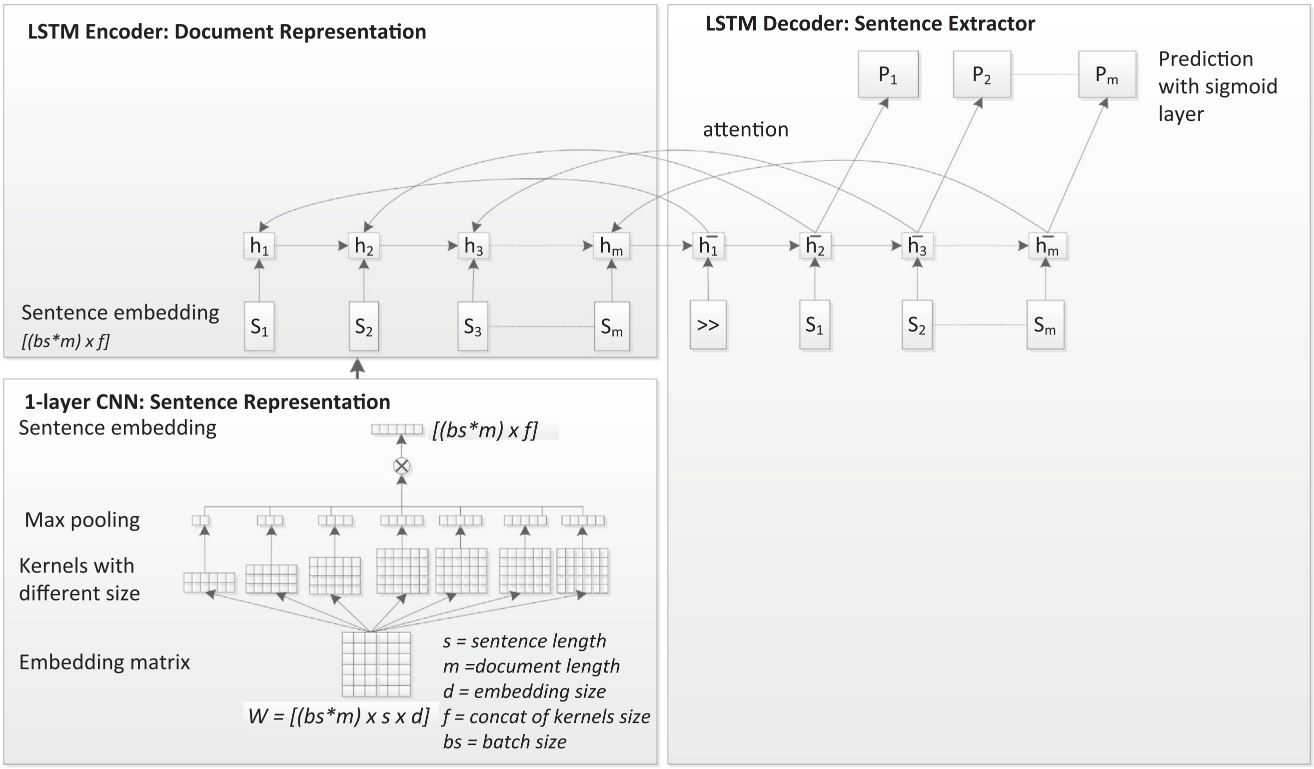

Fig. 1. Architecture of Sentence Extractor in NeuralSum, (). adapted from Cheng and Lapata, 2016

Where MLP is a multi-layer neural network that takes concatenation of $\bar{h}_t$ and h_t as an input. Although NeuralSum originally has word extractor in the architecture, it wasn't used in this study as it is more suitable for text generation.

3. Methodology

This section outlines the investigation of the development of bilingual word embedding based on mapping and joint methods for the pair of Indonesian–English languages. The procedure commenced with corpus pre-processing, i.e., the removal of non-alphanumeric characters and their conversion to lowercase. This was followed by the generation of word embedding for each language using SkipGram to create input for the VecMap method. Meanwhile, BiVec can directly train parallel corpus with or without an alignment process. These two methods yielded two aligned embeddings available for evaluation and implementation in NLP tasks. Therefore, this led to the assessment of both intrinsic and extrinsic evaluation. The bilingual lexicon induction task is used intrinsically, while extrinsic evaluation was adopted to extract text summarization as an NLP downstream task. The following subsection will further discuss in detail the dataset used, the bilingual word embedding method, and the evaluation procedure.

3.1. Generate and improve bilingual word embedding

The experiment was carried out with a widely used dataset in NLP study, Wikipedia and Opus Collection, in order to make it easier to compare with previous works. VecMap used Wikipedia as a monolingual corpus to generate the aligned embedding. Meanwhile, Opus Collection as a parallel corpus tends to be applied to both methods, namely VecMap and BiVec. The sizes of both corpora are shown in Tables 1 and 2.

Table 1
Monolingual Corpus.

Corpus	#Article	#Token
Wikipedia - EN	4.976.753	2.843.929.927
Wikipedia - ID	417.413	116.012.724

Creating a bilingual word embedding requires a large training corpus for each language $\{L_1, L_2\}$. This enables each i-th language to have a vocabulary $V_{L_i} = \{w_1, w_2, \cdots, w_{nL_i}\}$. Each of these is mapped to a shared vector space E, ensuring that the token with similar meaning also has the same value, enabling cross-lingual transfer learning.

This study implemented VecMap and BiVec by adopting the same settings since both utilize Word2Vec SkipGram to create word embedding. For VecMap, the Wikipedia corpus for each language was first trained independently by using the following parameter, Word2Vec SkipGram with 300 dimensions, negative sample 10, sub-sampling threshold 1e-5, five iterations, and maximum vocab size of 200,000. This parameter is in accordance with the experiment conducted by (Ormazabal et al., 2019). Next, the two embeddings from different languages were used to generate unsupervised shared spaces (Artetxe et al., 2018). Moreover, the experiment was carried out using a parallel corpus from Opus Collection. This aims to examine how it affects the VecMap alignment. The parallel corpus also has to be trained by BiVec under the same settings as VecMap, making it easier to compare the two approaches.

Based on the investigation of the BWE approaches, the performance of VecMap was further improved. Since VecMap learns cross-lingual embedding by mapping monolingual embedding without relying on parallel or comparable corpora, it is more suitable to be implemented in low-resource language than BiVec.

Table 2
Parallel Corpus.

Corpus	#docs	#sents	#EN token	#ID token
MultiCCAligned v1	1	27.0M	5.6G	180.0M
WikiMatrix v1	1	1.8M	1.0G	66.9M
CCAligned v1	541	27.0M	243.4M	233.9M
OpenSubtitles v2018	9827	9.7M	72.8M	60.9M
Tanzil v1	45	0.5M	8.5M	15.4M
Tatoeba v2020-11-09	3	10.0k	12.9M	91.8k
QED v2.0a	2219	0.4M	4.8M	3.8M
TED2020 v1	1740	0.2M	3.2M	2.7M
GNOME v1	1347	0.5M	2.7M	2.3M
bible-uedin v1	2	62.2k	1.8M	1.4M
Ubuntu v14.10	398	96.5k	0.6M	0.3M
GlobalVoices v2018	584	15.5k	0.4M	0.3M
KDE4 v2	125	15.1k	86.0k	91.1k
tico-19 v2020-10-28	1	3.1k	79.8k	74.5k
ELRC_2922 v1	1	2.7k	64.7k	59.4k
Total	16835	67.33M	6.95G	568.2M

VecMap requires unsupervised initialization, which is still perceived as a research challenge. Good initialization helps self-learning algorithms to find local optimum easily. Otherwise, it tends to lead to poor performance when randomly performed. VecMap's initial solution entails building a seed dictionary based on the assumption that two equivalent words in different languages should have similar distribution. However, initialization can also be carried out by building a shared vocabulary space. Irrespective of the lack of parallel corpus, training tends to be more effective by optimizing the vocabulary sharing of languages (Conneau et al., 2018).

The shared vocabulary is developed based on the assumption that tokens appearing in both monolingual corpus tend to have the same meaning (Camacho-Collados et al., 2020), such as numeric, names of persons, and places, alongside terms in foreign languages. The monolingual corpus from two languages was merged to find the shared vocabulary used to build shared spaces. Therefore, the same words in both languages tend to have similar embedding. The mapping process was only carried out on words that do not have any sharing information at all. This improvement method is shown in Fig. 2.

The improvement method in more detail is carried out as follows:

- Merge monolingual corpora for each language $\{L_1, L_2\}$
- Build vocabulary from each language and merged vocabulary $\{V_1, V_2, V\}$, where $V = V_1 \cup V_2$
- Generate embedding E from complete vocabulary V
- Relocate the vocabulary in order to put it according to its respective language, which produces three disjoint vocabulary sets $\{V_1, V_2, V_s\}$, where $V_s = V_1 \cap V_2$. This process is carried out to prevent oversharing in the shared space. The vocabulary is moved from shared space V_s to V_i when a word's frequency of occurrence in the -i language surpasses or equals the threshold. Given a vocabulary space V contains n words $V = \{w_1, w_2, \cdots, w_n\}$, the relocation is carried out by the following formula:

$$frek_{L_i}(w_j) = \frac{count_{L_i}(w_j)}{\sum count_{L_i}(w_j)} \tag{8}$$

$$V_i = \begin{cases} V_1, & frek_{L_1}(w) \geq \gamma \\ V_2, & frek_{L_2}(w) \geq \gamma \\ V_s, & otherwise \end{cases} \tag{9}$$

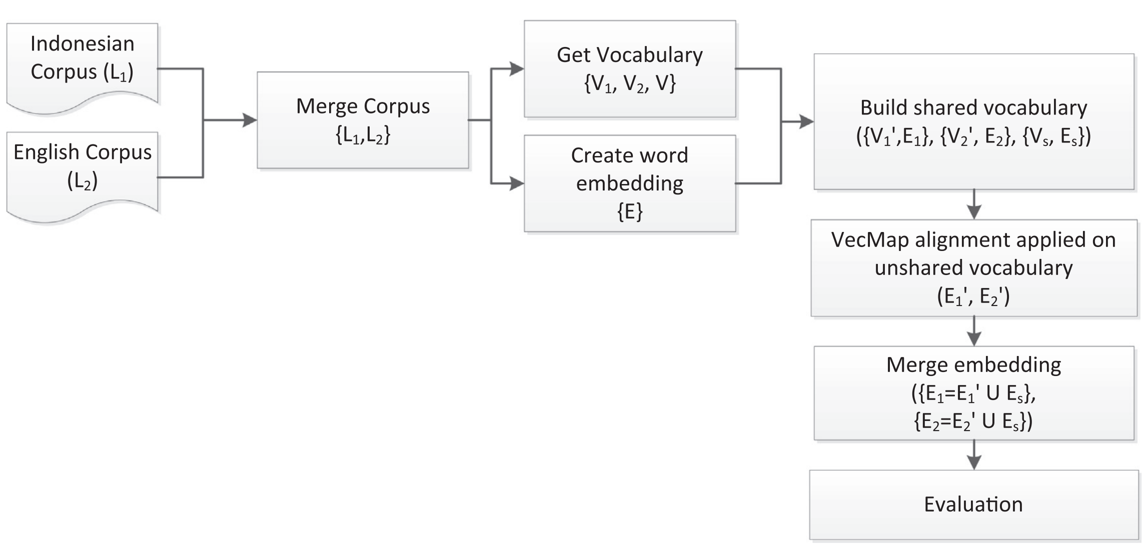

Fig. 2. The Improvement Method.

where $count_{L_i}(w_j)$ is the number word w_j in i- language, while $\sum count_{L_i}(w_j)$ is the number of words w_j in both languages. This process will result in three disjoint embeddings $\{E_1, E_2, E_s\}$.

The threshold is determined heuristically, within a value ranging between 0.6 and 0.9. This value is a trade-off between oversharing and shared space concept. The lower threshold γ causes the shared space to contain fewer words, thereby minimizing the possibility of oversharing. Therefore, the alignment process becomes increasingly important to obtain a better BWE. Otherwise, a high threshold will lead to oversharing due to a large number of words in the shared space.

- Perform VecMap alignment on vocabularies that are not in shared spaces to produce aligned embeddings $\{E_1', E_2'\}$
- Merge the aligned embeddings and embeddings in shared spaces to produce BWEs $\{E_1'', E_2''\}$

Since we proposed an improved Vecmap, we compare it to the existing version in Table 3, which outlines the current state of research.

3.2. Evaluation

The quality of the bilingual word embeddings was measured in terms of their performance when assessed with two NLP benchmarks, namely bilingual lexicon induction and text summarization. The first task intrinsically measures how well the bilingual embeddings inherently capture the relationships between the words in the source and target languages. Meanwhile, the second task assesses the ability of bilingual embedding to support model transfer across languages extrinsically.

3.2.1. Bilingual lexicon induction

The bilingual lexicon induction (BLI) is an important intrinsic evaluation in learning bilingual embeddings because it assesses how well they detect semantically similar word pairs across languages. It checks whether equivalent words in different languages are correctly translated in embedding space. The first step is to filter the reference bilingual lexicon such that it only contains entries existing in the BWE of the source and target languages. The lexicon pair will be used as the ground truth in this experiment once it is available. This number of ground truth is typically used as the denominator in BLI calculations (Conneau et al., 2018). However, following (Wang et al., 2020), the denominator in this paper cover the entire testing dataset. This is because some MUSE word pairings have the same translation, such as (lion, lion), (got, got). In the BWE of this study, these vocabularies would only be regarded as English. Likewise, the pairs of (pelangi, pelangi) will only be recognized as Indonesian. The identically spelled words in both languages should be allocated as shared vocabulary. Therefore, these conditions will still be included in the BLI calculation. However, the lexicon will be considered as OOV when the translation does not meet the criteria as a ground truth or has the same spelled word pair.

Table 3
Comparison of VecMap and Improved VecMap.

	VecMap	Improved VecMap
Embedding	Generated from a single language corpus	Generated from a merged corpus on both languages
Vocabulary	Allocated to the language based on the corpus of origin	Allocated to the language in accordance with the specified frequency threshold in a corpus
Alignment	All vocabulary in both languages	Only unshared vocabulary in both languages

In the word pair matching process, given a sequence of BWE's token in source language $s = \{s_1, s_2, \cdots, s_n\}$ and target language $t = \{t_1, t_2, \cdots, t_n\}$, an alignment between s and t is represented as (s_i, t_j). The alignment is determined by calculating its similarity using Nearest Neighbor (NN) or Cross-domain Similarity Local Scaling (CSLS). The unique token pair with the highest similarity score will be regarded as a translation. The result of this alignment is subsequently compared to the ground-truth. The evaluation process used in this task is precision-at-k (P@k) as the k-proportion of most similar words, but the current study simply employed P@1. Fig. 3 depicts the evaluation flow.

A bilingual dictionary derived from MUSE and WordNet was adopted as the ground truth, as shown in Table 4. MUSE provides bilingual dictionaries of 110 languages, and this data are widely used to assess the quality of its embedding, with a sample dictionary shown in Fig. 4. Meanwhile, WordNet was created by mapping the Indonesian and English synsets based on its ID. MUSE has split its dataset into training, validation, and testing data. In addition to testing data (MUSE-test), a complete dataset (MUSE-full) was used for the evaluation process since it is not involved in the bilingual word embedding training processes. Similarly, instead of only employing frequently used terms in WordNet (WordNet–freq), a complete dataset (WordNet–full) containing English and Indonesian synset pair was utilized.

3.2.2. Cross-Lingual learning for text summarization

Text summarization was applied as a downstream task to perform an extrinsic evaluation. The cross-lingual transfer aims to train the summarization model using bilingual embedding for source language L_1. In addition, the trained model on target language L_2 was evaluated by replacing the bilingual embedding with those of L_2. This task used English and Indonesian as the source and target languages, respectively. The architecture of the cross-lingual transfer learning for text summarization is illustrated in Fig. 5.

The NeuralSum (Cheng and Lapata, 2016) was used as a summarizer method in the architecture, as depicted in Fig. 4. It has an extractive summarization algorithm that utilizes word2vec embedding and encourages the use of simple cross-language mapping. The encoder generates document representation, while the decoder and attention mechanism select the salient sentences.

The NeuralSum model was trained on the DailyMail[2] corpus, containing 193,983 and 12,147 training as well as validation data, respectively, together with the English bilingual word embedding. It was subsequently evaluated on Liputan6 (Koto et al., 2020) and the IndoSum testing dataset as employed in (Wijayanti et al., 2021). The English bilingual word embedding is replaced with that of the Indonesian. The testing data consisted of 3,755 articles of IndoSum, and 10,972 articles of Liputan6. Furthermore, the following hyperparameters were applied, embedding dimensions 300, maximum epoch 20, seven kernels in CNN with different sizes (50,100,150,200,200,200,200), while the maximum summary length is three sentences for IndoSum and two for Liputan6, based on the average number of sentences in the summaries.

In the first scenario, the English summarization model was used to summarize the Indonesian text without fine-tuning (zero-shot transfer). Subsequently, 10%, 20%, 30%, and 40% additional training data in the target language were experimented with to fine-tune the summarization model, as shown in Table 5. However, the training and fine-tuning model was limited to five epochs due to limited computational resources.

The ROUGE (Rouges-1, 2, and L) scores were applied as an evaluation metric by comparing the results of the summarization methods with different scenarios. ROUGE (Recall-Oriented

[2] https://github.com/hpzhao/SummaRuNNer.

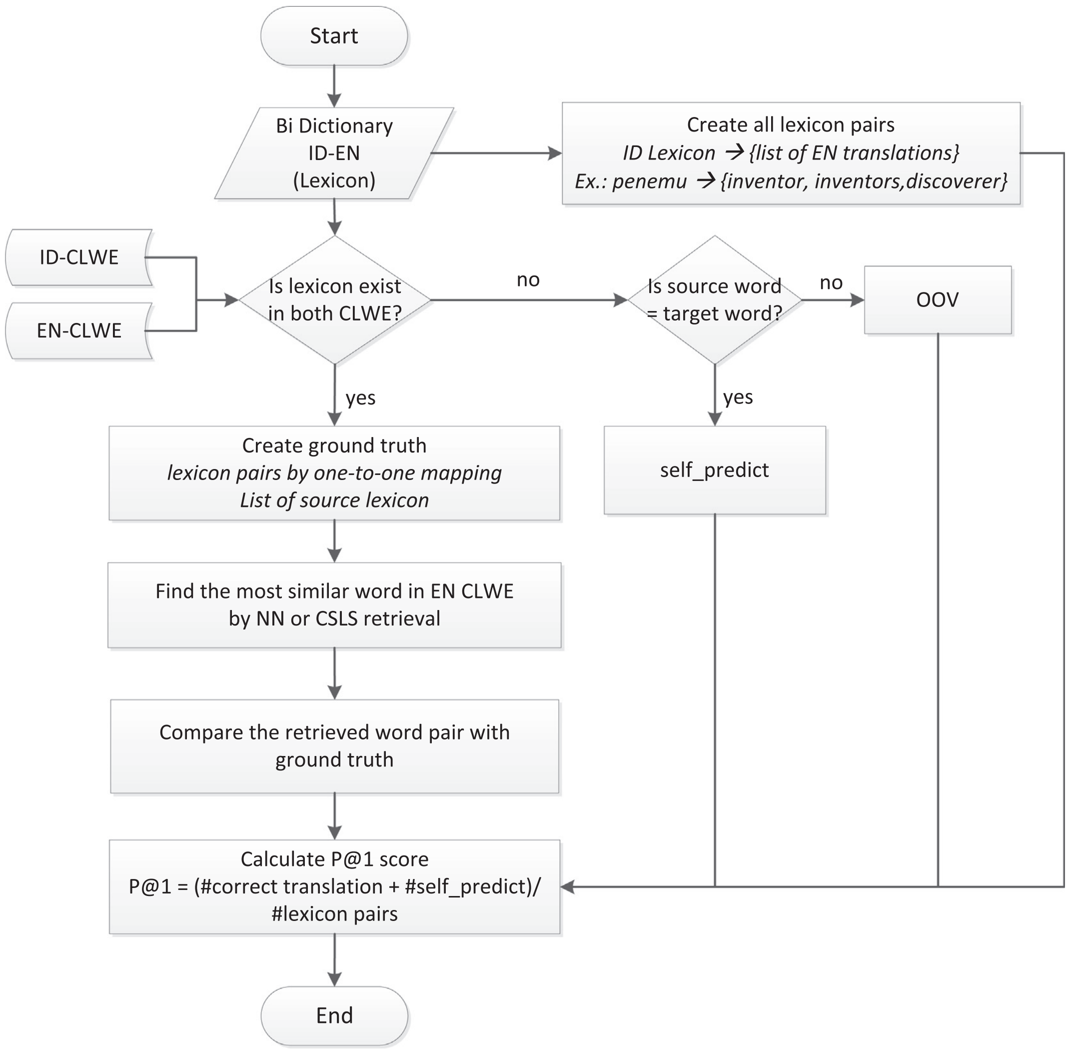

Fig. 3. Evaluation of Bilingual Lexicon Induction.

eksklusif	exclusive
eksklusif	exclusively
perselisihan	discord
perselisihan	strife
perselisihan	disputes
perselisihan	dispute
hipotesis	hypothesis
hipotesis	hypothetical

Fig. 4. The Example of Bilingual Dictionary.

Understanding for Gisting Evaluation) is a standard method for measuring the summarization procedure, which compares the summary results generated by a computer with the ideal one made by humans, known as candidate and reference summary (Lin, 2004). Meanwhile, ROUGE-N is the number of n-grams that overlap in the candidate and reference summaries. ROUGE-L is the longest common subsequence (LCS) ratio in the candidate and reference summaries.

Each ROUGE metric has Recall, Precision, and F1-score values. Similar to standard metrics used for classification or information retrieval (Jamaludin et al., 2022), recall represents the number of correct results divided by the number of results that should be returned. Meanwhile, precision shows the number of correct results divided by all the returned result. The harmonic mean of precision and recall is used to calculate the F1 score. In summarization task, given $w_{overlapped}$ as the number of overlapping words in candidate and reference summary, w_{ref} and w_{cand} are the numbers of terms in reference and candidate summaries. The following equations are used to calculate the metric:

$$Recall = \frac{w_{overlapped}}{w_{ref}} \tag{10}$$

$$Precision = \frac{w_{overlapped}}{w_{cand}} \tag{11}$$

$$F1score = \frac{2 \times Recall \times Precision}{(Recall + Precision)} \tag{12}$$

In this study, the words refer to unigram, bigram and LCS for ROUGEs-1, 2, and L, respectively. The abstract summary of each testing dataset serves as a reference in calculating the ROUGE score.

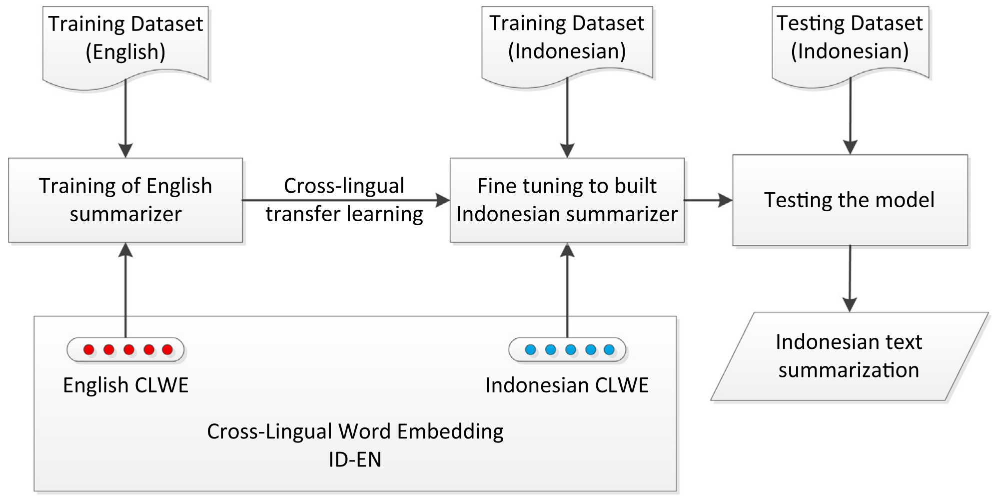

Fig. 5. The Architecture of Cross-Lingual Transfer Learning for Text Summarization.

Table 4
Bilingual Dictionary for Bilingual Lexicon Induction.

Corpus	#pair of words
MUSE – test	2453
MUSE – full	965518
WordNet – freq	11319
WordNet – full	284394

4. Experimental result and discussion

4.1. Bilingual lexicon induction (BLI)

This section commenced with the generation of the bilingual word embedding on monolingual corpus using the VecMap alignment. The results were subsequently compared to MUSE (Conneau et al., 2018), an existing bilingual word embedding widely used in various NLP studies. It supports multilingual word embedding for a number of languages, including Indonesian, which was also trained on Wikipedia corpus. The experiments result conducted on several testing data is shown in Table 6. It was discovered that VecMap outperforms MUSE on all given testing datasets.

The effect of the cross-lingual mapping approaches trained on monolingual and parallel corpus was subsequently investigated. First, the experimental results of VecMap, and BiVec trained on the monolingual (Wikipedia) and parallel corpus (OpenSubtitles v2018), respectively, were compared. Table 7 reveals that BiVec, which applied the joint method, outperforms VecMap on all testing datasets, especially when paired with CSLS retrieval. This result is slightly different from that of (Ormazabal et al., 2019), where CSLS only has minimal influence on the joint method.

It was assumed that both VecMap and BiVec still suffer from hubness problems in bilingual word embedding of English and Indonesian. Hubness is a problem that is often encountered in cross-lingual word embedding, where a few words (known as hubs) are the nearest neighbors of many other words in a high-dimensional space. It often degrades the accuracy of NN in various task and one of improvements have been made to mitigate the issue in BLI task is CSLS method (Conneau et al., 2018). CSLS relies on a local measure by penalizing the similarity scores of hubs, which in turn reduces the hubness level. This assumption is supported by the graph in Fig. 6, which indicates that the method using CSLS retrieval has lower hubness compared to the one using

Table 6
Bilingual Lexicon Induction of VecMap and Muse.

Dataset	**Method**	**P@1 NN**	**P@1 CSLS**
MUSE – test	MUSE	0.62	0.68
	VecMap	0.70	0.72
MUSE – full	MUSE	0.26	0.35
	VecMap	0.52	0.62
WordNet – freq	MUSE	0.30	0.33
	VecMap	0.31	0.34
WordNet – full	MUSE	0.30	0.35
	VecMap	0.42	0.46

Table 7
Bilingual Lexicon Induction of VecMap and BiVec.

Dataset	Method	P@1 NN	P@1 CSLS
MUSE-test	VecMap	0.70	0.72
	BiVec	0.78	0.85
MUSE-full	VecMap	0.52	0.62
	BiVec	0.76	0.84
WordNet-freq	VecMap	0.31	0.34
	BiVec	0.46	0.48
WordNet-full	VecMap	0.42	0.46
	BiVec	0.51	0.55

Table 5
Scenario of BWE Implementation in Text Summarization.

Scenario	#EN Train	#ID Train (IndoSum)	#ID Test (IndoSum)	#ID Train (Liputan6)	#ID Test (Liputan6)
Zero-shot	193983	0	3755	0	10972
ID10	193983	1351	3755	19388	10972
ID20	193983	2702	3755	38776	10972
ID30	193983	4053	3755	58164	10972
ID40	193983	5404	3755	77552	10972

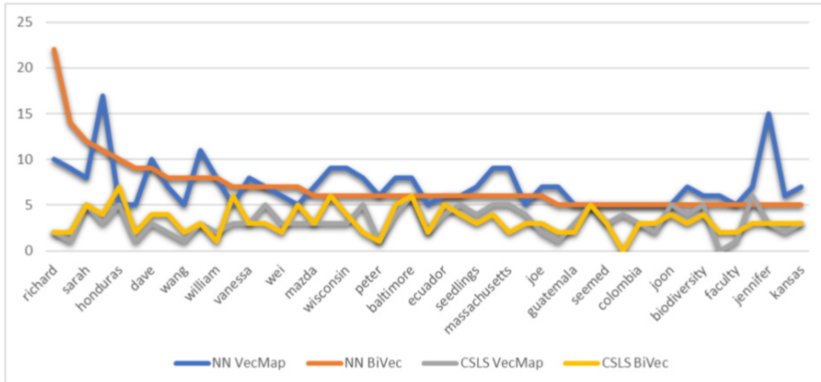

Fig. 6. Hubness in VecMap and BiVec.

Table 8
Bilingual Lexicon Induction of VecMap Trained on Monolingual vs Parallel Corpus.

Dataset	Corpus Type	P@1 NN	P@1 CSLS
MUSE – test	Monolingual	0.70	0.72
	Parallel	**0.86**	**0.87**
WordNet – freq	Monolingual	0.31	0.34
	Parallel	**0.44**	**0.44**

NN. Furthermore, the figure shows that the majority of hubness occurs in entities, such as people's names, locations, and others. Those entities should be translated into the same entities in different languages. Therefore, this study did not map these kinds of entities in the improved method because they will be in a shared space.

The joint method outperforms the mapping method since it is trained on supervised learning using a parallel corpus. The alignment signal is certainly more isomorphic than the corpus used in the mapping method. This is supported by the experimental results, as shown in Table 8. It indicates that VecMap trained on parallel corpus performs better than the simply trained on the monolingual type.

However, the VecMap approach, which only requires a monolingual corpus, was improved since obtaining a parallel corpus is quite challenging in low-resource language. By applying the improvement process previously displayed in Fig. 2, the results greatly outperformed the regular VecMap, as shown in Table 9. Although it did not surpass the BiVec, the improved VecMap that employed unsupervised alignment was relatively similar to BiVec results trained with supervised alignment. Furthermore, a low threshold ensures that the shared vocabulary space contains only a few words, thereby minimizing the possibility of oversharing. The alignment process is becoming increasingly important in terms of obtaining better bilingual word embedding.

Table 9
Bilingual Lexicon Induction of Improved VecMap.

Method	Threshold	P@1 CSLS
MUSE	–	0.68
BiVec	–	0.85
VecMap	–	0.72
Improved VecMap	0.9	0.75
Improved VecMap	0.8	0.81
Improved VecMap	0.7	0.83
Improved VecMap	0.6	0.84

Table 10
Extractive Summarization on IndoSum Dataset.

Method	R1-F	R2-F	RL-F
MonoNeuralSum	0.65	0.60	0.65
MonoBertSum-Ext (IndoBert)	0.67	0.60	0.66
Alignment with BiVec			
CLNeuralSum-ID0	0.64	0.57	0.61
CLNeuralSum-ID10	0.69	0.62	0.66
CLNeuralSum-ID20	0.69	0.62	0.66
CLNeuralSum-ID30	0.68	0.61	0.65
CLNeuralSum-ID40	0.67	0.60	0.64
Alignment with Improved VecMap			
CLNeuralSum-ID0	0.65	0.58	0.62
CLNeuralSum-ID10	0.68	0.61	0.65
CLNeuralSum-ID20	0.65	0.59	0.63
CLNeuralSum-ID30	0.67	0.61	0.65
CLNeuralSum-ID40	0.67	0.61	0.64

4.2. Cross-Lingual transfer Learning-Based extractive summarization

The extrinsic evaluation of the BiVec method and the improved Vecmap were analyzed in this section. This is because both methods have better intrinsic values than the regular VecMap alignment, as previously displayed in Table 9. Tables 10 and 11 display cross-lingual-based as well as monolingual-based summarization (monoNeuralSum and MonoBertSum-Ext) directly trained on the target language, i.e., IndoSum and Liputan6 datasets, respectively.

This experiment used monolingual summarizers, such as NeuralSum (Cheng and Lapata, 2016) and BertSum-Ext (Liu and Lapata, 2019) as the baseline. BertSum is a summarization method that adopts the BERT pre-trained model and outperforms NeuralSum. However, since the training process requires more computational resources, NeuralSum was applied to the cross-lingual summarization.

Table 11
Extractive Summarization on Liputan6 Dataset.

Method	R1-F	R2-F	RL-F
MonoNeuralSum	0.28	0.14	0.23
MonoBertSum-Ext (IndoBert)	0.38	0.21	0.35
Alignment with Improved VecMap			
CLNeuralSum-ID0	0.38	0.21	0.31
CLNeuralSum-ID10	0.39	0.21	0.31
CLNeuralSum-ID20	0.39	0.21	0.31
CLNeuralSum-ID30	0.39	0.21	0.31
CLNeuralSum-ID40	0.38	0.21	0.31

Article:
puluhan lembaga sosial di indonesia belum berstandar internasional .
seluruhnya masih dikelola dengan sederhana dan minim sumber daya manusia .
padahal lembaga sosial memiliki peran yang besar bagi peningkatan kapasitas hidup masyarakat di suatu negara .
hal tersebut diungkapkan rantala , perwakilan dari yayasan wafaa indonesia dalam forum peningkatan kapasitas organisasi atau civil community capacity building forum (cccb) dengan tema ' empowering civil society organization' , di jakarta , sabtu (9 / 9 / 2017) .
ditambahkan rantala , kegiatan yang bekerja sama dengan organisasi kemanusiaan nama foundation ini bertujuan meningkatkan kapasitas organisasi bagi lembaga sosial kemasyarakatan sesuai dengan standar internasional .
sebanyak 34 lembaga sosial mengikuti kegiatan yang digelar sejak jumat (8 / 9) lalu .
selain peningkatkan kapasitas keorganisasian , acara ini juga bertujuan membentuk jaringan dan mencari mitra baru di indonesia yang dapat bekerjasama dengan nama foundation , kata rantala .
rantala menyebutkan , peningkatan kapasitas organisasi ini sangat penting .
mengingat masih banyak organisasi berskala mikro yang dikelola dengan cara yang sangat sederhana dan memiliki sumber daya yang sangat terbatas .
karenanya , nama foundation berharap acara ini bisa bermanfaat untuk perwakilan lembaga yang hadir , tandasnya .
diketahui nama foundation didirikan pada tahun 2004 di malaysia dengan berkonsetrasi pada pengembangan sektor pendidikan dan pengembangan kapasitas organisasi .
nama foundation sendiri telah menelurkan beberapa kegiatan sosial di indonesia .
diantaranya peningkatan kapasitas di bidang pendidikan , peningkatan kapasitas buiding bagi konsultan sosial kemasyarkatan , dan peningkatan kapasitas pemimpin muda .

Reference Summary:
puluhan lembaga sosial di indonesia belum berstandar internasional .
seluruhnya masih dikelola dengan sederhana dan minim sumber daya manusia .
padahal lembaga sosial memiliki peran yang besar bagi peningkatan kapasitas hidup masyarakat di suatu negara .
hal tersebut diungkapkan rantala , perwakilan dari yayasan wafaa indonesia dalam civil community capacity building forum (cccb) dengan tema ' empowering civil society organization' , di jakarta , sabtu (9 / 9 / 2017) .

Candidate Summary:
puluhan lembaga sosial di indonesia belum berstandar internasional .
seluruhnya masih dikelola dengan sederhana dan minim sumber daya manusia .
padahal lembaga sosial memiliki peran yang besar bagi peningkatan kapasitas hidup masyarakat di suatu negara .

Fig. 7. Sample of summarization result. The underlined sentences are considered important sentences.

According to Table 9 and 10, the cross-lingual method produces a higher ROUGE score when fine-tuned with some Indonesian training data than the monolingual approach. Furthermore, by providing only 10% training data of the target language, the cross-lingual approach achieved a better ROUGE score than NeuralSum in a monolingual setting. This is also equivalent to BertSum result, although with smaller computational resources. An example of a summarization result is given in Fig. 7.

Moreover, the Friedman test was conducted in accordance with (Jamaludin et al., 2022) to validate the proposed cross-lingual transfer learning architecture using different text summarization methods across fine-tuning datasets. Given $\alpha = 0.05$ and degree of freedom $df = 2$, while the p-value for R1-F is 0.016 $(\chi^2 = 5.99147)$ and chi-square is 8.32. Therefore, the null hypothesis of equal performance given different fine-tuning datasets was rejected. Moreover, adjusting the amount of the dataset for fine tuning results in a different performance for the summary model.

4.3. Analysis

This section observes several typical errors made by Bilingual Lexicon Induction task and extractive models based on cross-lingual transfer model to better understand the tasks behavior.

4.3.1. Error analysis of bilingual lexicon induction

The lexicon pair resulting from word alignment method when tested on MUSE-test dataset was observed in order to perform error analysis in BLI task. This dataset was considered since the size

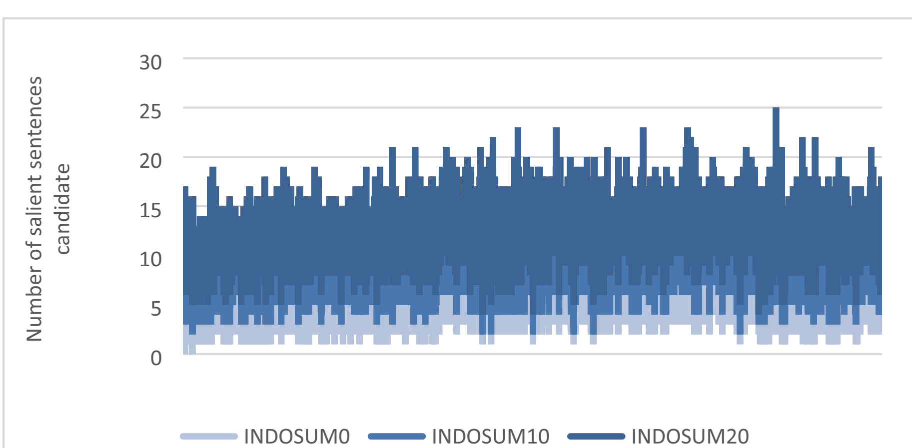

Fig. 8. Salient Sentences Candidate of IndoSum.

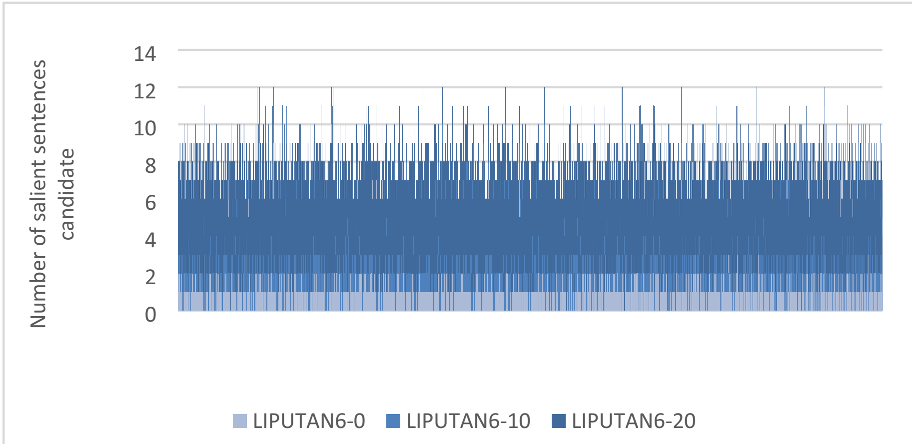

Fig. 9. Salient Sentences Candidate of Liputan6.

is relatively smaller compared to other test datasets, hence it is easier to observe. Some frequently occurring mistakes when using NN retrieval are as follows:

- Wrong translation, for example: “pernah” (ever) translated to “never”
- Lexical error, for example, “ofjustice” should be “justice”, “don” should be “don’t”
- Morphological differences, i.e., difference in plural, past or continuous form, while the dictionary only found singular form, for example, “makhluk” (creature) translated to “creatures”, “tertawa” (laugh) translated to “laughing”.
- Semantically related, for example, “berorientasi” (oriented) translated to “detail-oriented”
- Correctly translated, but not found in the dictionary, for example: “lewat” (pass) translated to “through” while “through” not found in ground truth.

Meanwhile, CSLS retrieval has fewer errors compared to that of NN. The errors found in CSLS typically occurred in the last three errors (morphological differences, semantically related, and lexicon that were correctly translated but not found in the dictionary). The morphological differences occur because there is no preprocessing technique, such as lemmatization or normalization in the word embedding generation since it lasts longer. Meanwhile, the limitation of the dictionary led to the lexical error and the last two errors because only the MUSE-test bilingual dictionary was used to assist the observation process.

4.3.2. Error analysis of extractive summaries

This section examines how adding fine-tuning datasets affects model behavior. As shown in Figs. 8 and 9, the addition of dataset in target language led to the model (NeuralSum) having more salient sentences as summary candidates. However, the n-sentences for the candidate were selected as the extractive summary is simply based on the position of the leadning sentence.

5. Conclusion

This study investigated the two bilingual word embedding methods, VecMap and BiVec, as enablers in the cross-lingual transfer learning concept. Interestingly, this concept solves problems in low-resource languages since an NLP model was developed by leveraging annotated data from high-resource languages. VecMap, as a mapping method, requires a monolingual corpus to create word embedding for each language independently and subsequently map the two embeddings into a shared vector space. Meanwhile, BiVec, as a joint method, requires a parallel corpus to find word representations from various languages simultaneously. Both methods were evaluated on bilingual lexicon induction and cross-lingual based-text summarization task.

According to the experiment results, bilingual word embedding based on the joint method outperformed the mapping method. This is because the joint method generated with a parallel corpus has a strong bilingual signal. However, due to the difficulty in obtaining a parallel corpus, there are other options than this approach for direct implementation in low-resource languages. Therefore, the VecMap’s performance was improved in such a way that it is at least comparable to or equal to that of Bivec’s. The improvements were made by constructing shared vocabulary spaces while alignment was performed on that of the unshared. This approach has been proven to improve VecMap performance by 16.6% when evaluated intrinsically. The extrinsic evaluation also shows that the Rouge score of improved VecMap is comparable to BiVec, and even better than monolingual training when just 10% of the data is provided in the target language.

Considering this potential result, the improvement method can be further enhanced by including NER or even classification to better allocate terms into shared spaces such that it is not only based on frequency.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Artetxe, M., Labaka, G., Agirre, E., 2018. A robust self-learning method for fully unsupervised cross-lingual mappings of word embeddings. In: ACL 2018 - 56th Annual Meeting of the Association for Computational Linguistics, Proceedings of the Conference (Long Papers). Association for Computational Linguistics, Melbourne, Australia, pp. 789–798. https://doi.org/10.18653/v1/p18-1073.

Budhi, G.S., Intan, R., R, S.R., 2007. Indonesian Automated Text Summarization. In: International Conference on Soft Computing, Intelligent System and Information Technology (ICSIIT). Bali, Indonesia.

Cai, Z., Lin, N., Ma, C., Jiang, S., 2019. Indonesian Automatic Text Summarization Based on A New Clustering Method in Sentence Level. In: International Conference on Big Data Engineering (BDE 2019). Association for Computing Machinery, Hongkong, pp. 30–35. https://doi.org/10.1145/3341620.3341626.

Camacho-Collados, J., Doval, Y., Martínez-Cámara, E., Espinosa-Anke, L., Barbieri, F., Schockaert, S., 2020. Learning cross-lingualword embeddings from Twitter via distant supervision. In: Proceedings of the 14th International AAAI Conference on Web and Social Media, ICWSM 2020, pp. 72–82.

Chandraseta, R., Khodra, M.L., 2019. Composing Indonesian Paragraph for Biography Domain using Extractive Summarization. In: 2019 International Conference on Advanced Informatics: Concepts, Theory, and Applications (ICAICTA 2019). IEEE, pp. 1–5. https://doi.org/10.1109/ICAICTA.2019.8904118.

Cheng, J., Lapata, M., 2016. Neural Summarization by Extracting Sentences and Words. In: The 54th Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, Berlin, Germany, pp. 484–494.

Conneau, A., Lample, G., Ranzato, M., Denoyer, L., Jégou, H., 2018. Word Translation Without Parallel Data, in: The 6th International Conference on Learning Representations (ICLR). Microtome Publishing, Vancouver, Canada. https://doi.org/10.1111/j.1540-4560.2007.00543.x.

Devlin, J., Chang, M.-W., Lee, K., Toutanova, K., 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In: NAACL-HLT 2019. Association for Computational Linguistics, Minneapolis, Minnesota, pp. 4171–4186.

Doval, Y., Camacho-Collados, J., Espinosa-Anke, L., Schockaert, S., 2020. On the robustness of unsupervised and semi-supervised cross-lingual word embedding learning. In: LREC 2020 - 12th International Conference on Language Resources and Evaluation, Conference Proceedings. European Language Resources Association (ELRA), Marseille, France, pp. 4013–4023.

Goodfellow, I.J., Pouget-Abadie, J., Mirza, M., Xu, B., David Warde-Farley, S.O., Courville, A., Bengio, Y., 2014. Generative Adversarial Nets. Adv. Neural Inf. Process. Syst. 2672–2680. https://doi.org/10.1016/j.vetimm.2013.08.005.

Gouws, S., Bengio, Y., Corrado, G., 2015. BilBOWA: Fast Bilingual Distributed Representations without Word Alignments. In: The 32nd International Conference on Machine Learning.

Gouws, S., Søgaard, A., 2015. Simple task-specific bilingual word embeddings. In: The 2015 Annual Conference of the North American Chapter of the ACL. Association for Computational Linguistics, Denver, Colorado, pp. 1386–1390.

Gunawan, D., Pasaribu, A., Rahmat, R.F., Budiarto, R., 2017. Automatic Text Summarization for Indonesian Language Using TextTeaser. In: IOP Conference Series: Materials Science and Engineering. IOP Publishing. https://doi.org/10.1088/1742-6596/755/1/011001.

Halim, K., Novianus Palit, H., Tjondrowiguno, A.N., 2020. Penerapan Recurrent Neural Network untuk Pembuatan Ringkasan Ekstraktif Otomatis pada Berita Berbahasa Indonesia. J. Infra 8, 221–227.

Herdiyeni, Y., Ridha, A., Adisantoso, J., 2012. Text feature weighting for summarization of document Bahasa Indonesia using genetic algorithm. Int. J. Comput. Sci. Issues 9, 1–6.

Jamaludin, S.Z.M., Romli, N.A., Kasihmuddin, M.S.M., Baharum, A., Mansor, M.A., Marsani, M.F., 2022. Novel logic mining incorporating log linear approach. J. King Saud Univ. – Comput. Inf. Sci. 34, 9011–9027. https://doi.org/10.1016/j.jksuci.2022.08.026.

Koto, F., Lau, J.H., Baldwin, T., 2020. Liputan6: A Large-scale Indonesian Dataset for Text Summarization. In: The 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing. Association for Computational Linguistics, Suzhou, China, pp. 598–608.

Koto, F., 2016. A publicly available Indonesian corpora for automatic abstractive and extractive chat summarization. In: Proceedings of the 10th International Conference on Language Resources and Evaluation, LREC 2016. pp. 801–805.

Kurniawan, K., Louvan, S., 2018. IndoSum: A New Benchmark Dataset for Indonesian Text Summarization. In: The International Conference on Asian Language Processing (IALP 2018). IEEE, Bandung, Indonesia, pp. 215–220. https://doi.org/10.1109/IALP.2018.8629109.

Lample, G., Conneau, A., 2019. Cross-lingual Language Model Pretraining. In: The 33rd Conference on Neural Information Processing Systems (NeurIPS 2019), Vancouver, Canada.

Lin, C., 2004. ROUGE: A Package for Automatic Evaluation of Summaries. Workshop on Text Summarization Branches Out. Association for Computational Linguistics, Barcelona, Spain.

Liu, Y., Lapata, M., 2019. Text Summarization with Pretrained Encoders. In: The 2019 Conference on Empirical Methods in Natural Language Processing and The 9th International Joint Conference on Natural Language Processing. Association for Computational Linguistics, Hong Kong, China, pp. 3728–3738.

Luong, T., Pham, H., Manning, C.D., 2015. Bilingual Word Representations with Monolingual Quality in Mind. In: The 1st Workshop on Vector Space Modeling for Natural Language Processing. Association for Computational Linguistics, Denver, Colorado, pp. 151–159. https://doi.org/10.3115/v1/W15-1521.

Marie, B., Fujita, A., 2019. Unsupervised joint training of bilingual word embeddings. In: ACL 2019 - 57th Annual Meeting of the Association for Computational Linguistics, Proceedings of the Conference. Association for Computational Linguistics, pp. 3224–3230. https://doi.org/10.18653/v1/p19-1312.

Maryana, F., Kurniawati, A., Agusten, D., 2018. Term frequency method for automated text summarization application of Indonesian news article. In: The 3rd International Conference on Informatics and Computing (ICIC 2018). IEEE, pp. 1–7. https://doi.org/10.1109/IAC.2018.8780443.

Mikolov, T., Le, Q.V., Sutskever, I., 2013. Exploiting Similarities among Languages for Machine Translation.

Ormazabal, A., Artetxe, M., Labaka, G., Soroa, A., Agirre, E., 2019. Analyzing the Limitations of Cross-lingual Word Embedding Mappings. In: The 57th Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, pp. 4990–4995.

Peng, X., Zheng, Y., Lin, C., Siddharthan, A., 2021. Summarising Historical Text in Modern Languages. In: The 16th Conference of the European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers. Association for Computational Linguistics, Online.

Pikuliak, M., Šimko, M., Bieliková, M., 2021. Cross-lingual learning for text processing: a survey. Expert Syst. Appl. 165. https://doi.org/10.1016/j.eswa.2020.113765.

Ruder, S., Vulić, I., Søgaard, A., 2019. A survey of cross-lingual word embedding models. J. Artif. Intell. Res. 65. https://doi.org/10.17863/CAM.30462.

Sabet, A., Gupta, P., Cordonnier, J.B., West, R., Jaggi, M., 2019. Robust cross-lingual embeddings from parallel sentences, arXiv.

Sabuna, P.M., Setyohadi, D.B., 2017. Summarizing Indonesian text automatically by using sentence scoring and decision tree. In: 2017 2nd International Conferences on Information Technology, Information Systems and Electrical Engineering (ICITISEE 2017). IEEE, pp. 1–6. https://doi.org/10.1109/ICITISEE.2017.8285473.

Silvia, R., Aprilia, V.R., Suhartono, D., Wongso, R., Meiliana, 2014. Summarizing text for indonesian language by using latent dirichlet allocation and genetic algorithm. Int. Conf. Electr. Eng. Comput. Sci. Informatics 1, 148–153. https://doi.org/10.11591/eecsi.1.364.

Søgaard, A., Ruder, S., Vulic, I., 2018. On the Limitations of Unsupervised Bilingual Dictionary Induction. In: The 56th Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, Melbourne, Australia, pp. 778–788.

Tardan, P.P., Erwin, A., Eng, K.I., Muliady, W., 2013. Automatic Text Summarization Based on Semantic Analysis Approach for Documents in Indonesian Language. In: 2013 International Conference on Information Technology and Electrical Engineering (ICITEE 2013). IEEE, pp. 47–52. https://doi.org/10.1109/ICITEED.2013.6676209.

Vuli, I., Moens, M.-F., 2016. Bilingual distributed word representations from document-aligned comparable data. J. Artif. Intell. Res. 55, 953–994.

Wang, Z., Xie, J., Xu, R., Yang, Y., Neubig, G., Carbonell, J., 2020. Cross-Lingual Alignment Vs Joint Training: A Comparative Study And A Simple Unified Framework. In: The 8th International Conference on Learning Representations.

Wijayanti, R., Khodra, M.L., Widyantoro, D.H., 2021. Indonesian Abstractive Summarization using Pre-Trained Model. In: 3rd 2021 East Indonesia Conference on Computer and Information Technology, EIConCIT 2021. pp. 79–84. https://doi.org/10.1109/EIConCIT50028.2021.9431880.

Žagar, A., Robnik-Šikonja, M., 2020. Cross-lingual approach to abstractive summarization, arXiv.

Zhang, M., Liu, Y., Luan, H., Sun, M., 2017. Adversarial Training for Unsupervised Bilingual Lexicon Induction. In: The 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics, Vancouver, Canada, pp. 1959–1970. https://doi.org/10.18653/v1/P17-1179.